

IT4IT Mapping & Reflective Summary

Project: ProposalAI — AI Technical Proposal Generator

1. Overview

ProposalAI is an AI-assisted decision support tool designed for SME clients to automatically generate technical proposals (CCTV, Networking, POS, IT infrastructure).

Instead of manually calculating bill of materials and costs, the system processes user input and produces a structured proposal including:

- Bill of Materials (BoM)
- Cost estimation
- Executive summary
- Risk insights
- Monitoring data

The system follows the IT4IT framework to demonstrate that it is not just a prototype, but a viable business-oriented solution.

2. IT4IT Lifecycle Mapping

Strategy to Portfolio (S2P)

The S2P stage defines the business problem and context.

In this project, S2P is implemented through the input sidebar, where the user specifies:

- Client name
- Solution type (e.g. CCTV, Networking, POS)
- Area (m²)

- Number of floors
- Number of users
- Budget constraints
- Risk tolerance

This stage ensures that the system understands the real-world business scenario before generating any solution.

The problem addressed:

> SMEs lack expertise and time to design technical infrastructure proposals efficiently.

AI is a suitable solution because:

- It reduces manual engineering effort
- Standardizes proposal generation
- Provides faster decision-making

Requirement to Deploy (R2D)

R2D represents how the solution is designed and implemented.

Instead of manually coding the full business logic, the system is structured into modular components:

- Domain calculators:
 - CCTV calculation
 - Networking estimation
 - POS system generation
- Scoring engine:
 - Evaluates cost, risk, scalability
- Proposal generator:
 - Combines all modules into a final structured output

The architecture is agent-guided, meaning:

- The system design and logic were defined through prompts
- The AI agent assisted in generating and refining the implementation

This reflects a shift from coding → system orchestration and prompt engineering

Request to Fulfill (R2F)

R2F describes how the user consumes the system.

In ProposalAI, the interface is a Streamlit dashboard, which provides:

- Real-time proposal generation
- KPI overview (cost, components, score)
- Bill of Materials table
- Executive summary
- Insights panel

The interface is designed for non-technical business users, allowing them to:

- Quickly generate proposals
- Understand costs and risks
- Make decisions without engineering knowledge

Detect to Correct (D2C)

D2C focuses on monitoring and reliability.

The system includes an AI Monitoring panel, which tracks:

- API calls (or simulated calls)
- Token usage
- Cost estimation
- Response latency
- AI confidence score
- Error flags (e.g. low confidence input)

Additionally, the Insights system detects:

- Missing data
- Risk conditions
- Budget inconsistencies

This ensures that:

- The system remains transparent
- Users can trust the generated outputs
- Errors can be identified and corrected

3. Reflection

Working with an AI agent instead of writing all logic manually significantly changed the development process.

The main challenges were:

- Designing correct system structure before implementation
- Writing precise prompts to guide the AI
- Debugging generated code rather than writing it

Unlike traditional programming, the focus shifted to:

- Architecture thinking
- Decomposition of the problem into modules
- Iterative refinement through prompts

One of the hardest parts was ensuring consistency:

- Sometimes outputs were incomplete
- Some UI elements required multiple refinements
- Design and usability needed iterative improvement

However, this approach demonstrated that:

> Managing AI is closer to product architecture than coding.

4. Conclusion

ProposalAI successfully demonstrates all four IT4IT value streams:

- S2P → Business problem definition through inputs
- R2D → AI-assisted system design and modular logic
- R2F → User-facing dashboard delivering value
- D2C → Monitoring and validation mechanisms

The system is not only functional but also scalable and adaptable to different SME scenarios.